

\* Rank 0 tensor. \* Rank 1 tensor. \* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = G_{dxy}^{(1)}$$

\* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & G_{xy} & 0 \\ -G_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$G_{xy} = Q_{pz}^{(1)}$$

\* Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{yzx} & 0 \\ 0 & 0 & G_{yzx} \end{bmatrix}$$

$$G_{yzx} = G_{f3}^{(1)}$$

\* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & -G_{yzx} - G_{zxy} \end{bmatrix}$$

$$G_{yzx} = 3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

$$G_{zxy} = -3Q_{du}^{(1)} + Q_{dv}^{(1)}$$

\* Rank 3 tensor (a).

$$\begin{bmatrix} G_{yzx} & 0 & 0 \\ 0 & G_{zxy} & 0 \\ 0 & 0 & G_{xyz} \end{bmatrix}$$

$$G_{yzx} = -Q_{du}^{(2)} + Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{zxy} = -Q_{du}^{(2)} - Q_{dv}^{(2)} + Q_s^{(1)}$$

$$G_{xyz} = 2Q_{du}^{(2)} + Q_s^{(1)}$$

\* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & G_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & G_{yyxy} \\ 0 & 0 & 0 & 0 & 0 & G_{zzxy} \\ 0 & 0 & 0 & 0 & G_{yzzx} & 0 \\ 0 & 0 & 0 & G_{yzzx} & 0 & 0 \\ G_{xxxy} & G_{yyxy} & G_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{gaz}^{(1)} - G_{gbz}^{(1)} \\
G_{yyxy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{gaz}^{(1)} - G_{gbz}^{(1)} \\
G_{zzxy} &= G_{dxy}^{(1)} + 2G_{gbz}^{(1)} \\
G_{yzzx} &= G_{dxy}^{(2)} + 2G_{gbz}^{(1)}
\end{aligned}$$

\* Rank 4 tensor (ssa).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & G_{xxxy} \\
0 & 0 & 0 & 0 & 0 & G_{yyxy} \\
0 & 0 & 0 & 0 & 0 & G_{zzxy} \\
0 & 0 & 0 & 0 & G_{yzzx} & 0 \\
0 & 0 & 0 & -G_{yzzx} & 0 & 0 \\
-G_{xxxy} & -G_{yyxy} & -G_{zzxy} & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= G_{dxy}^{(3)} - 2Q_{faz}^{(1)} + 2Q_{fbz}^{(1)} + 2Q_{pz}^{(1)} \\
G_{yyxy} &= G_{dxy}^{(3)} + 2Q_{faz}^{(1)} + 2Q_{fbz}^{(1)} - 2Q_{pz}^{(1)} \\
G_{zzxy} &= G_{dxy}^{(3)} - 4Q_{fbz}^{(1)} \\
G_{yzzx} &= -4Q_{faz}^{(1)} - Q_{pz}^{(1)}
\end{aligned}$$

\* Rank 4 tensor (aas).

$$\begin{bmatrix}
0 & G_{yzzx} & 0 \\
G_{yzzx} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$G_{yzzx} = 2G_{dxy}^{(4)}$$

\* Rank 4 tensor (aaa).

$$\begin{bmatrix}
0 & G_{yzzx} & 0 \\
-G_{yzzx} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$G_{yzzx} = -Q_{pz}^{(2)}$$

\* Rank 4 tensor (sa).

$$\begin{bmatrix}
0 & 0 & G_{xxxy} \\
0 & 0 & G_{yyxy} \\
0 & 0 & G_{zzxy} \\
0 & G_{yzzx} & 0 \\
G_{zxxy} & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= -2G_{dxy}^{(5)} - Q_{faz}^{(2)} + Q_{fbz}^{(2)} + Q_{pz}^{(3)} \\
G_{yyxy} &= 2G_{dxy}^{(5)} - Q_{faz}^{(2)} - Q_{fbz}^{(2)} + Q_{pz}^{(3)} \\
G_{zzxy} &= 2Q_{faz}^{(2)} + Q_{pz}^{(3)} + 2Q_{pz}^{(4)} \\
G_{yzzx} &= -G_{dxy}^{(5)} - Q_{faz}^{(2)} - Q_{fbz}^{(2)} + Q_{pz}^{(4)} \\
G_{zxxy} &= G_{dxy}^{(5)} - Q_{faz}^{(2)} + Q_{fbz}^{(2)} + Q_{pz}^{(4)}
\end{aligned}$$

\* Rank 4 tensor (as).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & G_{yzzx} & 0 \\
0 & 0 & 0 & G_{zxxy} & 0 & 0 \\
G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{yzzx} &= G_{dxy}^{(6)} - Q_{faz}^{(3)} + Q_{fbz}^{(3)} + Q_{pz}^{(6)} \\
G_{zxyz} &= -G_{dxy}^{(6)} - Q_{faz}^{(3)} - Q_{fbz}^{(3)} + Q_{pz}^{(6)} \\
G_{xyxx} &= -2G_{dxy}^{(6)} - Q_{faz}^{(3)} + Q_{fbz}^{(3)} + Q_{pz}^{(5)} \\
G_{xyyy} &= 2G_{dxy}^{(6)} - Q_{faz}^{(3)} - Q_{fbz}^{(3)} + Q_{pz}^{(5)} \\
G_{xyzz} &= 2Q_{faz}^{(3)} + Q_{pz}^{(5)} + 2Q_{pz}^{(6)}
\end{aligned}$$

\* Rank 4 tensor (s).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & G_{xxxy} & 0 & 0 & G_{xxyx} \\
0 & 0 & 0 & 0 & 0 & G_{yyxy} & 0 & 0 & G_{yyyyx} \\
0 & 0 & 0 & 0 & 0 & G_{zzxy} & 0 & 0 & G_{zzyx} \\
0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\
0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\
G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{xxxy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} + G_{gaz}^{(1)} - G_{gbz}^{(1)} - 2Q_{faz}^{(1)} - Q_{faz}^{(2)} + 2Q_{fbz}^{(1)} + Q_{fbz}^{(2)} + 2Q_{pz}^{(1)} + Q_{pz}^{(3)} \\
G_{xxyx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} + 2G_{dxy}^{(5)} + G_{gaz}^{(1)} - G_{gbz}^{(1)} - 2Q_{faz}^{(1)} + Q_{faz}^{(2)} + 2Q_{fbz}^{(1)} - Q_{fbz}^{(2)} + 2Q_{pz}^{(1)} - Q_{pz}^{(3)} \\
G_{yyxy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} + 2G_{dxy}^{(5)} - G_{gaz}^{(1)} - G_{gbz}^{(1)} + 2Q_{faz}^{(1)} - Q_{faz}^{(2)} + 2Q_{fbz}^{(1)} - Q_{fbz}^{(2)} - 2Q_{pz}^{(1)} + Q_{pz}^{(3)} \\
G_{yyyyx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} - G_{gaz}^{(1)} - G_{gbz}^{(1)} + 2Q_{faz}^{(1)} + Q_{faz}^{(2)} + 2Q_{fbz}^{(1)} + Q_{fbz}^{(2)} - 2Q_{pz}^{(1)} - Q_{pz}^{(3)} \\
G_{zzxy} &= G_{dxy}^{(1)} + G_{dxy}^{(3)} + 2G_{gbz}^{(1)} + 2Q_{faz}^{(2)} - 4Q_{fbz}^{(1)} + Q_{pz}^{(3)} + 2Q_{pz}^{(4)} \\
G_{zzyx} &= G_{dxy}^{(1)} + G_{dxy}^{(3)} + 2G_{gbz}^{(1)} - 2Q_{faz}^{(2)} - 4Q_{fbz}^{(1)} - Q_{pz}^{(3)} - 2Q_{pz}^{(4)} \\
G_{yzzx} &= G_{dxy}^{(2)} - G_{dxy}^{(5)} + 2G_{gbz}^{(1)} - 4Q_{faz}^{(1)} - Q_{faz}^{(2)} - Q_{fbz}^{(2)} - Q_{pz}^{(1)} + Q_{pz}^{(4)} \\
G_{yzxz} &= G_{dxy}^{(2)} + G_{dxy}^{(5)} + 2G_{gbz}^{(1)} - 4Q_{faz}^{(1)} + Q_{faz}^{(2)} + Q_{fbz}^{(2)} - Q_{pz}^{(1)} - Q_{pz}^{(4)} \\
G_{zxyz} &= G_{dxy}^{(2)} + G_{dxy}^{(5)} + 2G_{gbz}^{(1)} + 4Q_{faz}^{(1)} - Q_{faz}^{(2)} + Q_{fbz}^{(2)} + Q_{pz}^{(1)} + Q_{pz}^{(4)} \\
G_{zxzy} &= G_{dxy}^{(2)} - G_{dxy}^{(5)} + 2G_{gbz}^{(1)} + 4Q_{faz}^{(1)} + Q_{faz}^{(2)} - Q_{fbz}^{(2)} + Q_{pz}^{(1)} - Q_{pz}^{(4)} \\
G_{xyxx} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{dxy}^{(3)} + G_{gaz}^{(1)} - G_{gbz}^{(1)} + 2Q_{faz}^{(1)} - 2Q_{fbz}^{(1)} - 2Q_{pz}^{(1)} \\
G_{xyyy} &= G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{dxy}^{(3)} - G_{gaz}^{(1)} - G_{gbz}^{(1)} - 2Q_{faz}^{(1)} - 2Q_{fbz}^{(1)} + 2Q_{pz}^{(1)} \\
G_{xyzz} &= G_{dxy}^{(1)} - G_{dxy}^{(3)} + 2G_{gbz}^{(1)} + 4Q_{fbz}^{(1)}
\end{aligned}$$

\* Rank 4 tensor (a).

$$\begin{bmatrix}
0 & 0 & 0 & 0 & G_{yzzx} & 0 & 0 & G_{yzxz} & 0 \\
0 & 0 & 0 & G_{zxyz} & 0 & 0 & G_{zxzy} & 0 & 0 \\
G_{xyxx} & G_{xyyy} & G_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

$$\begin{aligned}
G_{yzzx} &= 2G_{dxy}^{(4)} + G_{dxy}^{(6)} - Q_{faz}^{(3)} + Q_{fbz}^{(3)} - Q_{pz}^{(2)} + Q_{pz}^{(6)} \\
G_{yzxz} &= -2G_{dxy}^{(4)} + G_{dxy}^{(6)} - Q_{faz}^{(3)} + Q_{fbz}^{(3)} + Q_{pz}^{(2)} + Q_{pz}^{(6)} \\
G_{zxyz} &= 2G_{dxy}^{(4)} - G_{dxy}^{(6)} - Q_{faz}^{(3)} - Q_{fbz}^{(3)} + Q_{pz}^{(2)} + Q_{pz}^{(6)} \\
G_{zxzy} &= -2G_{dxy}^{(4)} - G_{dxy}^{(6)} - Q_{faz}^{(3)} - Q_{fbz}^{(3)} - Q_{pz}^{(2)} + Q_{pz}^{(6)} \\
G_{xyxx} &= -2G_{dxy}^{(6)} - Q_{faz}^{(3)} + Q_{fbz}^{(3)} + Q_{pz}^{(5)} \\
G_{xyyy} &= 2G_{dxy}^{(6)} - Q_{faz}^{(3)} - Q_{fbz}^{(3)} + Q_{pz}^{(5)} \\
G_{xyzz} &= 2Q_{faz}^{(3)} + Q_{pz}^{(5)} + 2Q_{pz}^{(6)}
\end{aligned}$$

\* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & G_{xxxy} & 0 \\ G_{yyyyx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{zzxy} & 0 \\ G_{xxyx} & 0 & 0 \\ G_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & G_{xyyy} & 0 \\ 0 & 0 & \frac{G_{xxxxy}}{2} - \frac{G_{xxyyx}}{2} + G_{xyyy} - G_{yyyx} + G_{zzxy} \end{bmatrix}$$

$$G_{xxxy} = G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} + G_{gaz}^{(1)} - G_{gbz}^{(1)}$$

$$G_{yyyyx} = G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} - 2G_{dxy}^{(5)} - G_{gaz}^{(1)} - G_{gbz}^{(1)}$$

$$G_{zzxy} = G_{dxy}^{(1)} + G_{dxy}^{(3)} + 2G_{gbz}^{(1)}$$

$$G_{xxyx} = G_{dxy}^{(1)} + 2G_{dxy}^{(2)} + G_{dxy}^{(3)} + 2G_{dxy}^{(5)} + G_{gaz}^{(1)} - G_{gbz}^{(1)}$$

$$G_{yzzx} = G_{dxy}^{(2)} - G_{dxy}^{(5)} + 2G_{gbz}^{(1)}$$

$$G_{xyyy} = G_{dxy}^{(1)} + 2G_{dxy}^{(2)} - G_{dxy}^{(3)} - G_{gaz}^{(1)} - G_{gbz}^{(1)}$$